Amogh Parab

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SUMMARY

Mathematician and Lean4 engineer with expertise in type theory, categorical algebra, and formal verification • **Contributed to Mathlib** and currently formalizing over 200 advanced mathematical problems to **support AI training pipelines at Harmonic • PhD in Mathematics from The Ohio State University** with a thesis on coherence of categorical groups, bridging rigorous abstract theory with practical formalization workflows • **Experienced in building reusable Lean4 libraries** and verifying complex algebraic and topological structures • Passionate about applying formal methods to the development of trustworthy AI systems.

SKILLS

- Formal Methods: Lean4 (Mathlib, Verified-zkEVM-CompPoly), dependent type theory, formal verification, theorem proving
- Languages: Python (NumPy, SciPy, Pandas, Matplotlib, Scikit-learn), LaTeX, Julia
- Mathematics: Category Theory, Algebraic Topology, Abstract Algebra, Type Theory, Numerical Methods
- OS and Platforms: Bash, GitHub, Windows, Linux, Jupyter Notebook, Visual Studio Code

WORK EXPERIENCE

Harmonic — **Remote**
Lean Expert (part-time) — Sep. 2025 – Present

- **Formalized over 200 advanced mathematics problems** in Lean4, spanning a broad range of topics, including algebra, geometry, and number theory to support the development of an advanced mathematical reasoning model.
- **Contributed to open-source formalization projects**, including Mathlib, Google DeepMind's Formal Conjectures, and Verified-zkEVM-CompPoly, expanding the frontier of machine-verifiable mathematics.
- Designed and implemented a Euclidean Geometry library in Lean4 to streamline formalization workflows.

The Ohio State University — **Columbus, Ohio**
Graduate Teaching Associate / Lecturer — Aug. 2020 – Present

- Created and presented over **150 hours of mathematics content**, incorporating visual demonstrations to enhance student engagement, with an average teaching score of 4.7/5.
- Led Engineering Mathematics recitations, conducted office hours, proctored exams, and prepared teaching materials for undergraduate engineering students.

PROJECT EXPERIENCE

Rewriting Systems in Lean4 (GitHub) — **Columbus, Ohio**
The Ohio State University — Aug. 2023 – Jan. 2025

- **Designed and implemented an inductive data type in Lean4** to serve as a rewriting system for multiplicative words, a core component of the categorical group structures in my PhD thesis.
- Formally verified fundamental algebraic properties, including **associative concatenation and the universal property**; authored over **2,000 lines of Lean4 code** and **20+ pages of technical documentation in LaTeX.**

Coherence of Categorical Groups (GitHub) — **Columbus, Ohio**
The Ohio State University — Aug. 2023 – Present

- Extended Mac Lane's coherence theorem for monoidal categories to categorical groups, providing proofs of three distinct coherence statements; the core theoretical contribution of my PhD thesis.
- **Formalized the core algebraic structure of categorical groups** in Lean4; actively working to contribute the formalization to Mathlib.

Formalizing Homotopy Lifting Property (GitHub) — **Berkeley, California**
SLMath (formerly MSRI) — Jun. 2023

- **Collaborated with Mathlib maintainers** at SLMath to port the Lean mathematics library from Lean3 to Lean4 during a two-week formalization summer school.
- **Formalized the uniqueness of the homotopy lifting property** in Lean4, writing over 300 lines of code; presented findings alongside a team of six at the closing session.

Data Science Bootcamp (Project 1: [GitHub]) (Project 2: [GitHub]) **Remote**
Participant at the Erdős Institute Sep. 2024 – Nov. 2024

- Predicted results of Valorant online matches with **67% accuracy using XGBClassifier** on selected key features from 160 engineered features; applied **grid search for hyperparameter tuning.**
- Developed a predictive model for car prices with an R^2 value of 0.88 using XGBoost in a team of four, **achieving a 22% improvement** over the baseline model; utilized SHAP values to identify key feature importance.

Creep Evolution in FCC crystalline structure **College Station, Texas**
Texas A&M University (Internship) May 2017 – Jul. 2017

- Analyzed stress-strain relationship of exposed creep on external boundary conditions using ABAQUS CAD software.

Quasi-Solution Approach to Non-Linear Differential Equations **Columbus, Ohio**
The Ohio State University (Internship) Aug. 2016 – Jan. 2016

- Provided analytic solution to the Blasius boundary condition (Fluid Dynamics) using Maple software. Used the Banach fixed-point theorem to show the existence of the solution.

PUBLICATIONS

- "Coherence and Symmetrization of Categorical Groups", PhD thesis, The Ohio State University, 2025
- "Gröbner bases of Rational Normal Curves", Proceedings of the COCOA Workshop, IIT Gandhinagar, 2019

GRANTS AND AWARDS

- Special Graduate Assignment for Thesis Research, The Ohio State University, 2023
- Bronze Medal for the best performance in the core courses of Mathematics, IIT Gandhinagar, 2019
- Dean's List, IIT Gandhinagar, Spring 2015, Fall 2015, Spring 2016

EDUCATION

The Ohio State University **Columbus, Ohio**
PhD in Mathematics (Cumulative GPA: 3.99/4) Aug. 2025
Thesis: Coherence and Symmetrization of Categorical Groups
Coursework: Mathematics of Data Science, Abstract Algebra, Analysis, Lie Theory, Topological Quantum Field Theory

Indian Institute of Technology Gandhinagar **Gandhinagar, India**
BTech in Mechanical Engineering, MSc in Mathematics (Cumulative GPA: 3.90/4) Jul. 2019
Master's Thesis: On Riemann-Roch Theorem
Coursework (Master's): Linear Algebra, Numerical Methods, Ordinary and Partial Differential Equations, Group Theory
Coursework (BTech): Quantum Mechanics, Non-Linear Dynamics

LEADERSHIP EXPERIENCE

Directed Reading Program **Columbus, Ohio**
Program Coordinator, The Ohio State University Aug. 2021 – May 2025

- Organized the pairing process of more than 1**00 undergraduate mathematics enthusiasts** with experienced graduate mentors to explore topics of mutual interest.
- **Mentored five students over two years** in advanced mathematical concepts beyond the standard curriculum; introduced formal theorem proving via Lean's **Natural Number Game.**

Team LaTeX **Gandhinagar, India**
Co-founder, IIT Gandhinagar Aug. 2017 – May 2018

- **Co-founded a LaTeX club** at IIT Gandhinagar to train students and researchers in professional scientific typesetting; organized week-long workshops attended by hundreds of students and faculty.